Dynamic Treatment Effect Estimation with Interactive Fixed Effects and Short Panels

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https://kylebutts.com/files/JMP.pdf

Motivation

Empirical Application

General Identification Result

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Treatment is often targeted to places/units based on their economic trends:

- New apartment construction (Asquith, Mast, and Reed, 2021; Pennington, 2021)
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Standard difference-in-differences assumption of parallel counterfactual trends is *implausible*

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- New apartment construction
  - $\rightarrow$ Changing preferences for walkable neighborhoods
- Walmart entry
  - $\rightarrow$ Growing employment increases disposable income
- Place-based policies
  - $\rightarrow$ Decline of manufacturing hurting manufacturing hubs

Modeling differential trends

This paper models differential trends using a **factor model** that is popular in the finance/macroeconomics literature:

- Each time period there is a set of *unobservable* macroeconomic shocks that are common across units
- Units vary based on their *unobservable* baseline characteristics in how impacted they are by the shocks

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This paper models differential trends using a **factor model** that is popular in the finance/macroeconomics literature:

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Allows treated units to be on trends *based on macroeconomic shocks*

This paper

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We propose a **class of imputation-style treatment effect estimators** under a **factor model** that is fixed-T consistent under treatment-effect heterogeneity

- Our 'imputation' style estimator explicitly estimates the untreated potential outcome, $y_{it}(0)$, in the post-treatment periods

  $\rightarrow$ Same strategy as synthetic control and imputation estimator in difference-in-differences context (Borusyak, Jaravel, and Spiess, 2024)

This paper

We provide a general formula to take a $\sqrt{N}$-consistent estimate of the macroeconomic shocks to be used to generate consistent treatment effect estimates

- Unlocks a large econometric literature on factor model estimation and incorporates it into causal inference methods
  - $\rightarrow$ e.g. principal components, quasi-long differencing, common-correlated effects, etc.

▸ Quasi-long Differencing Details ▸ Principal Components Details ▸ Common Correlated Effects Details

Current approaches

Synthetic-control style estimators

There are many estimators for treatment effects under factor models:

1. Synthetic control (Abadie, 2021)
2. Matrix Completion (Athey et al., 2021; Bai and Ng, 2021)
3. Imputation Estimators (**Xu_2017**; Gobillon and Magnac, 2016)

None of these are valid in fixed-T settings or require non-autocorrelated errors

Importance of fixed-T estimatiors

In general, adapting large-T estimators to 'small'-T settings can be undesirable:

- In short-panels, you *over-fit* on noise and get bad estimates (Abadie, Diamond, and Hainmueller, 2010; Ferman and Pinto, 2021)
- Even if you have a large number of pre-periods, structural changes to the economy can make far-away pre-periods uninformative (e.g. the 2008 recession) (Abadie, 2021)

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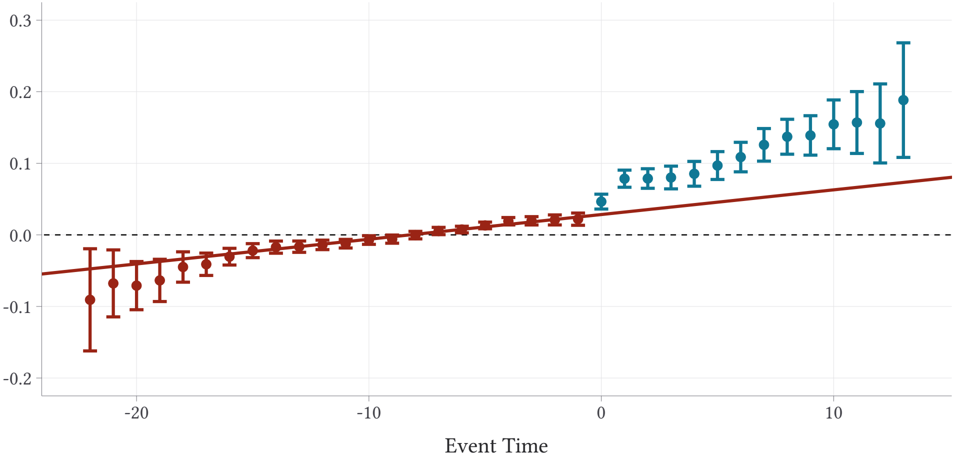
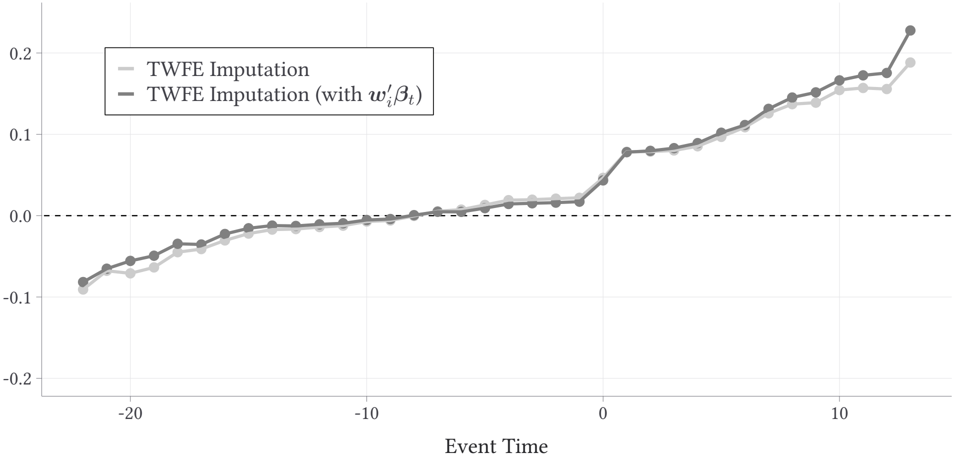
Impact of new Walmart entry on $\log$ retail employment

TWFE Imputation estimates



Impact of new Walmart entry on $\log$ retail employment

Including $w_i \beta_t$ in TWFE imputation



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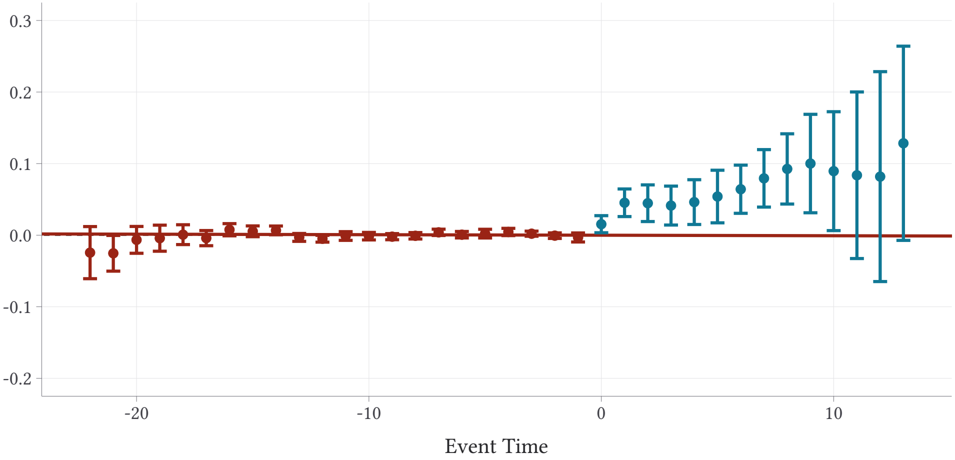
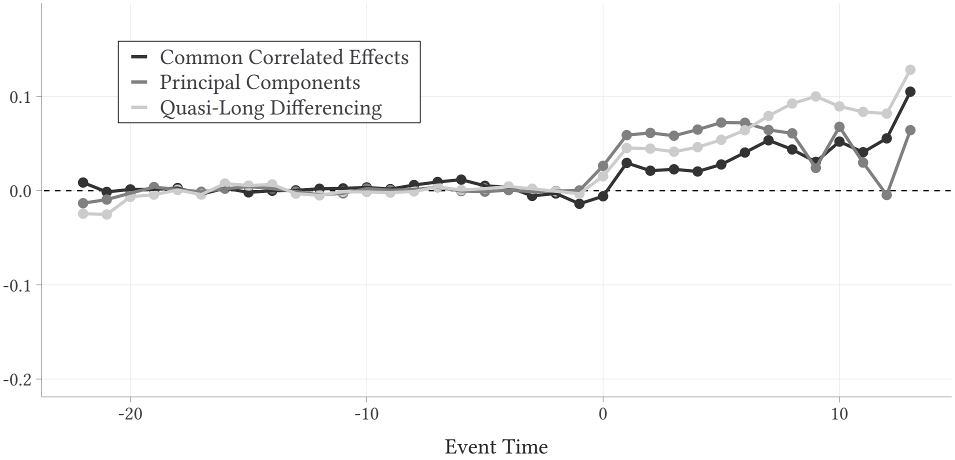
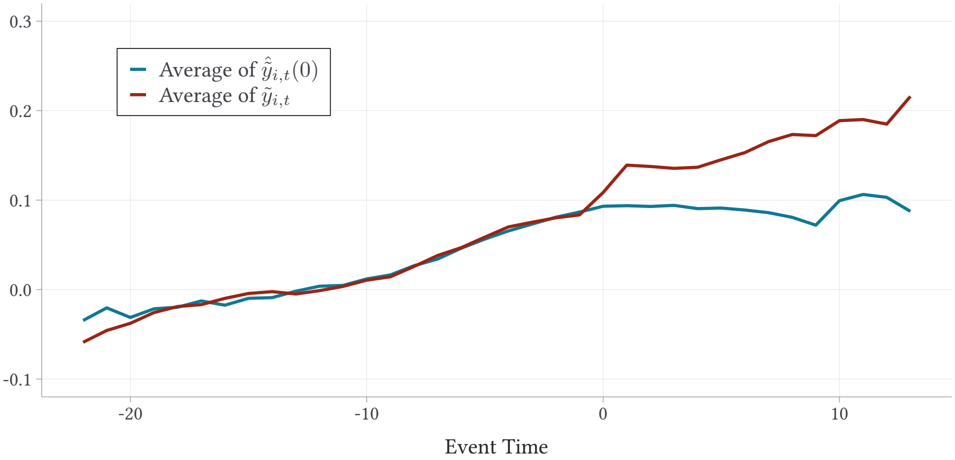
Quasi-long differencing estimates using w_i as instruments



Can use many different factor-model estimators



Imputation estimator allows synthetic control style plots



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Model

We observe an iid panel of observations denoted by unit $i \in \{1, \ldots, N\}$ and by time period $t \in \{1, \ldots, T\}$.

Untreated potential outcomes are given by a factor model:

$$y_{it}(0) = \sum_{r=1}^{p} f_{t,r} * \gamma_{i,r} + u_{it} \tag{1}$$

- $f_{t,r}$ is the r-th **factor** (macroeconomic shock) at time t.
- $\gamma_{i,r}$ is unit i's **factor loading** (exposure) to the r-th factor.

Two-way Fixed Effect vs. Factor Model

The factor model is a generalization of the TWFE model. If $\boldsymbol{f}_t = (\lambda_t, 1)'$ and $\boldsymbol{\gamma}_i = (1, \mu_i)'$, then (1) becomes the TWFE model:

$$y_{it}(0) = \boldsymbol{f}_t' \gamma_i + u_{it} = \lambda_t + \mu_i + u_{it}$$

Since TWFE is the work-horse model used by applied researchers, later we show to add unit and time fixed-effects back in and how to test sufficiency of TWFE model.

Treatment Effects

For now, assume there is a single treatment that turns on in some period $T_0 + 1$. Define D_i to be a dummy to denote which units receive treatment and d_{it} to equal 1 when treatment is active.

- We assume a large number of treated and control units $N_1 = \sum_i D_i$ and $N_0 = \sum_i D_i$ are non-vanishing as N grows

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We are interested in event-study style average treatment effects. For each t, we define

$$\text{ATT}_t \equiv \mathbb{E}[y_{it}(1) - y_{it}(0) \mid D_i = 1],$$

where $y_{it}(0)$ is the (unobserved) untreated potential outcome.

Assumptions

Factor model

Assumption: Selection into Treatment

Untreated potential outcomes are given by

$$y_{it}(0) = \boldsymbol{f}_t' \boldsymbol{\gamma}_i + u_{it}.$$

where $\mathbb{E}[u_{it} \mid \boldsymbol{\gamma}_i, D_i] = \mathbb{E}[u_{it} \mid \boldsymbol{\gamma}_i] = 0$ for all t.

- Relaxes parallel trends by allowing units to enter treatment based on exposure to macroeconomic shocks

Assumptions

Additional assumptions

Assumption: Arbitrary Treatment Effects

Treatment effects are left unrestricted (besides having finite moments):

$$\tau_{it} = y_{it}(1) - y_{it}(0)$$

Assumption: No Anticipation

Whenever $d_{it} = 0$, the observed $y_{it} = y_{it}(0)$.

ATT_t Identification

For a given t, our selection into treatment assumption implies:

$$\begin{aligned}
\text{ATT}_t &\equiv \mathbb{E}[y_{it}(1) - y_{it}(0) \mid D_i = 1] \\
&= \mathbb{E}\left[y_{it}(1) - \left(\boldsymbol{f}_t' \boldsymbol{\gamma}_i + u_{it}\right) \mid D_i = 1\right] \\
&= \mathbb{E}[y_{it}(1) \mid D_i = 1] - \boldsymbol{f}_t' \, \mathbb{E}[\boldsymbol{\gamma}_i \mid D_i = 1]
\end{aligned}$$

Insight: Estimating each γ_i requires large-T

- We only need to estimate $\mathbb{E}[\gamma_i \mid D_i = 1]$ which is possible in small-T settings

ATT$_t$ Identification

Suppose we observed the $T \times p$ matrix of factors, $\boldsymbol{F}$. Let 'pre' denote the rows corresponding to $t \leq T_0$. Then for $t > T_0$,

$$
\begin{aligned}
&\mathbb{E}\left[y_{it} - \boldsymbol{f}_t' (\boldsymbol{F}_{\text{pre}}' \boldsymbol{F}_{\text{pre}})^{-1} \boldsymbol{F}_{\text{pre}}' \underbrace{\boldsymbol{y}_{i,\text{pre}}}_{\boldsymbol{F}_{\text{pre}} \gamma_i + \boldsymbol{u}_{i,\text{pre}}} \mid D_i = 1 \right] \\
&\quad = \mathbb{E}\left[y_{it} - \boldsymbol{f}_t' \gamma_i \mid D_i = 1 \right] \\
&\quad = \mathbb{E}[y_{it} - y_{it}(0) \mid D_i = 1] = ATT_t,
\end{aligned}
$$

where the first equality comes from $\mathbb{E}[u_{it} \mid D_i = 1] = 0$.

Estimation of $\boldsymbol{F}$

$$ATT_t = \mathbb{E}\big[y_{it} - \boldsymbol{f}_t'(\boldsymbol{F}_{\text{pre}}'\boldsymbol{F}_{\text{pre}})^{-1}\boldsymbol{F}_{\text{pre}}'\boldsymbol{y}_{i,\text{pre}} \mid D_i = 1\big]$$

Consistency requires $\sqrt{n}$-consistent estimation of the column-span of the factors $\boldsymbol{F}$

- This brings in a large literature on factor model estimation to causal-inference methods

Conclusion

Provide general identification results for ATTs under linear factor models.

- Generalizes the two-way fixed effect model and is estimable in short-T settings
- Can use multiple estimators for the factor space.
  - $\rightarrow$ Brown et al. (2023) for CCE.

Implement quasi-long-differencing estimator

- Find results on Walmart's affect on local labor markets similar to Basker (2005).

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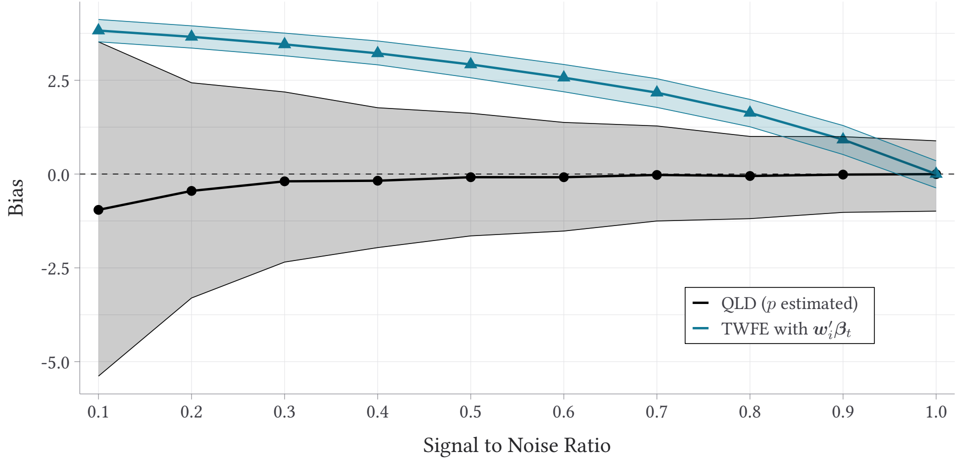
Figure: TWFE model with noisy proxy variable $w_i = \gamma_i + v_i$



Back

Intuition of Factor Model

The intuition is very similar to that of the construction of a shift-share variable:

$$z_{it} = \sum_{r=1}^{p} f_{t,r} * \gamma_{i,r}$$

- The $p \times 1$ vector $\boldsymbol{f}_t$ is the set of *'macroeconomic'* shocks (shifts) that all units experience
- γ_i is an individuals *exposure* (shares) to the shocks

The difference being that **we do not observe** the variables γ_i and $\boldsymbol{f}_t$ (like we don't observe fixed effects)

Removing additive effects

We consider the residuals after within-transforming

$$\tilde{y}_{it} = y_{it} - \overline{y}_{0,t} - \overline{y}_{i,pre} + \overline{y}_{0,pre},$$

where

$$\overline{y}_{i,pre} = \frac{1}{T_0}\sum_{t=1}^{T_0} y_{it}, \qquad \overline{y}_{0,t} = \frac{1}{N_0}\sum_{i=1}^{N}(1 - D_i)y_{it}, \qquad \overline{y}_{0,pre} = \frac{1}{T_0}\sum_{t=1}^{T_0} y_{0,t}$$

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Test for TWFE Model

The ATTs are identified by the modified TWFE transformation if

$$\mathbb{E}[\boldsymbol{\gamma}_i \mid D_i] = \mathbb{E}[\boldsymbol{\gamma}_i] \tag{2}$$

For $t > T_0$

$$\mathbb{E}[\tilde{y}_{it} \mid D_i = 1] = \mathbb{E}[\tau_{it} \mid D_i = 1] = \tau_t$$

- Says TWFE is sufficient even if there are factors, so long as exposure to these factors are the same between treated and control group.

In the paper, we provide tests for (2) under the quasi-long differencing identification strategy

▸ Back

Factor Identification

We cannot identify $\boldsymbol{F}$ or γ_i separately from one another.

- Rotation problem means we can only identify $A\boldsymbol{F}$ for some matrix A.

Consider some estimator $\boldsymbol{F}(\theta)$ such that the true factor matrix $\boldsymbol{F} \in \text{col}(\boldsymbol{F}(\theta))$

- **Examples:** common correlated effects, principal components, quasi-differencing.

Then using $\boldsymbol{F}(\theta)$ in place of $\boldsymbol{F}$ still identifies ATT_t.

- $\boldsymbol{f}_t(\boldsymbol{F}_{\text{pre}}'\boldsymbol{F}_{\text{pre}})^{-1}\boldsymbol{F}_{\text{pre}}$ is invariant to rotating by any invertible matrix $A\boldsymbol{F}$.

Quasi-long Differencing Details

The quasi-long differencing method Ahn, Lee, and Schmidt (2013) normalize the factors:

$$\boldsymbol{F}(\boldsymbol{\theta}) = \begin{pmatrix} -\boldsymbol{I}_p \\ \boldsymbol{\Theta} \end{pmatrix}$$

- Recall, this normalization does not impact imputation.

Quasi-differencing transformation: $\boldsymbol{H}(\boldsymbol{\theta}) = [\boldsymbol{\Theta}, \boldsymbol{I}_{(T-p)}]$. For all θ, we have

$$\boldsymbol{H}(\boldsymbol{\theta})\boldsymbol{F}(\boldsymbol{\theta}) = 0$$

Factor Identification

This transformation creates a set of moments:

$$\mathbb{E}[\boldsymbol{w}_i \boldsymbol{H}(\boldsymbol{\theta}) \boldsymbol{y}_i \mid D_i = 0]$$

- W_i is a $(T - p) \times w$ matrix of instruments ($w \geq p$).
- W_i must be exogenous after removing factors.
- $\hat{\boldsymbol{\theta}}$ is Fixed-T consistent.

Assumption: General framework for factor estimators

There exists a $q \times 1$ vector of paramters $\boldsymbol{\theta}$ and a $T \times m$ function $\boldsymbol{F}(\boldsymbol{\theta})$ such that the following conditions hold:

(i) For some full-rank matrix $\boldsymbol{A}$, $\boldsymbol{F}(\boldsymbol{\theta})\boldsymbol{A} = \boldsymbol{F}$ where $\text{Rank}(\boldsymbol{F}(\boldsymbol{\theta})) = m < T_0$

(ii) There is a $s \times 1$ vector of moment functions $\boldsymbol{g}_{i\infty}(\boldsymbol{\theta})$ such that

$$\mathbb{E}[\boldsymbol{g}_{i\infty}(\boldsymbol{\theta}) \mid G_i = \infty] = \mathbf{0}$$

(iii) Let $\boldsymbol{D}_\infty = \mathbb{E}[\nabla_{\boldsymbol{\theta}} \boldsymbol{g}_{i\infty}(\boldsymbol{\theta}) \mid G_i = \infty]$. Then $\text{Rank}(\boldsymbol{D}_\infty) = q$.

(iv) $\mathbb{E}[\boldsymbol{g}_{i\infty}(\boldsymbol{\theta})\boldsymbol{g}_{i\infty}(\boldsymbol{\theta})' \mid G_i = \infty]$ is positive definite.

Selection of p

The paper assumes that the number of factors, p, is known. Methods for selecting p depend on the factor estimator

- For the quasi-long differencing estimator, Ahn, Lee, and Schmidt (2013) provide a procedure for asymptotically determining p (given valid instruments):
  - $\rightarrow$ Start with $p = 0$ and estimate model. Calculate a J-statistic for the GMM model fit. If you reject null, then increase p.
  - $\rightarrow$ Continue this until you fail to reject the null. This asymptotically selects the correct p
- For principal-components, **Xu_2017**, discuss a similar selection procedure
- For the common-correlated effects estimator, Brown, Butts, and Westerlund (2023) show that you only need the number of covariates to be larger than p.

Principal Components Details

The principal component analysis takes the $T \times T$ matrix:

$$\mathbb{E}\left[\boldsymbol{y}_i \boldsymbol{y}_i' \mid D_i = 0\right]$$

- Use the never-treated sample to estimate the covariance matrix.

The first p eigenvectors of the PC-decomposition will serve as the estimate of $\boldsymbol{F}$.

- Consistently spans the column space of $\boldsymbol{F}$ if $t \to \infty$ or if error term u_{it} is independent within i.

▸ PC Strategy

Common Correlated Effects Details

The common-correlated effects model assumes there are $K \geq p$ covariates that each take the form of:

$$x_{k,it} = \sum_{r=1}^{p} \xi_{i,r}^{k} f_{t,r} + \nu_{it}^{k},$$

where $f_{t,r}$ are the same factor shocks as the original outcome model.

The factor proxies $\boldsymbol{F}_t$ are formed as cross-sectional averages of x for the never-treated sample:

$$\hat{\boldsymbol{F}}_t' = \left(\mathbb{E}[x_{1,it} \mid D_i = 0], \ldots, \mathbb{E}[x_{K,it} \mid D_i = 0] \right)$$

▸ CCE Strategy

Application: China's WTO Accension

Lu and Yu (2015) analyze the pro-competitive effects of China's World Trade Organization accession

Their headline figure is that after ascension industry-level price dispersion declined in China by about 5% in the years following

- The provide *suggestive* evidence that this occured via a decline in TFP dispersion *and* a decline in the marginal cost dispersion.

Application: China's WTO Accension

Our CCE estimator in Brown, Butts, and Westerlund (2023) introduces a technique that allows to decompose treatment effects into a mediated effect (via changing the value of x_{it}) and a direct effect (occuring via other chanels)

- Allow for treatment to effect control variables x_{it} (Caetano et al., 2022)

Figure: Impact of China's WTO Asscension on Markup Dispersion

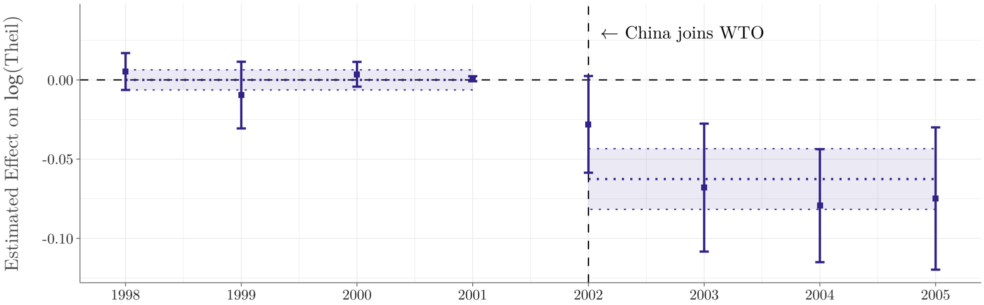

Figure: Mediated Effect via A Decline in TFP Dispersion